12TH CHEMISTRY · CHAPTER 32 · SOLVED EXERCISE

Medicine, Agriculture and Industry

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS

I) What does the TD_{50} value in the TI formula represent?

(a) Dose effective for 100 % of people.
(b) Lowest dose that causes harm MTC.
(c) Lowest dose that works MEC
(d) Dose causing a toxic effect in 50 % of people.

Answer: (d) Dose causing a toxic effect in 50 % of people. — TD_{50} is the dose that causes a toxic effect in 50% of the people who take it.

II) What is the key structural feature for penicillin's antibiotic activity?

(a) A carboxylic acid group
(b) A benzene ring
(c) A thiazolidine ring
(d) A four-membered beta-lactam ring

Answer: (d) A four-membered beta-lactam ring — The beta-lactam ring opens and binds permanently to the bacterial enzyme.

III) How do opiates like morphine reduce pain?

(a) They destroy pain signals.
(b) They bind to opioid receptors, blocking pain signals.
(c) They come from the opium poppy.
(d) They build new cell walls in the brain.

Answer: (b) They bind to opioid receptors, blocking pain signals. — Key (opiate) and lock (receptor): pain signals are blocked from reaching the brain.

IV) Which N-P-K nutrient promotes protein synthesis and chlorophyll?

(a) Nitrogen
(b) Phosphorus
(c) Potassium
(d) Sulphur

Answer: (a) Nitrogen — Nitrogen is a component of amino acids and of chlorophyll.

V) How does acid rain primarily damage soil?

(a) It increases soil pH.
(b) It leaches essential nutrients and releases toxic Al^{3+}.
(c) It adds excess nitrogen.
(d) It directly destroys plant roots.

Answer: (b) It leaches essential nutrients and releases toxic Al^{3+}. — H^+ ions displace Ca^{2+}, Mg^{2+} and K^+ and make Al^{3+} soluble.

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VI) What is a primary effect of climate change on agriculture?

(a) Reduced atmospheric CO_2
(b) Increased crop yields globally
(c) Reduced yields due to heat stress and erratic rainfall
(d) Fewer extreme weather events

Answer: (c) Reduced yields due to heat stress and erratic rainfall — Heat stress, droughts and floods reduce growth and cause crop failure.

VII) In Pakistan, rock salt (halite) is a key raw material for producing:

(a) Fertilizers
(b) Caustic soda and chlorine
(c) Cement
(d) Petrochemicals

Answer: (b) Caustic soda and chlorine — NaCl from Khewra gives soda ash, caustic soda and chlorine.

VIII) Ammonia gas in industry is prepared by:

(a) Contact process
(b) Haber-Bosch process
(c) Solvay process
(d) Refining process

Answer: (b) Haber-Bosch process — N_2 and H_2 combine over an iron catalyst.

XI) Plastics, synthetic fibres, and synthetic rubber are primary products of the:

(a) Cement industry
(b) Petrochemical industry
(c) Cosmetics industry
(d) Food production industry

Answer: (b) Petrochemical industry — They are made from the small molecules obtained by cracking crude oil.

The ninth MCQ is numbered "XI)" in the textbook (it should be IX); the printed number is kept.

X) What is the primary purpose of ventilation systems in a chemical facility?

(a) To control storage temperature
(b) To provide clear emergency exits
(c) To assist in labeling chemicals
(d) To remove toxic fumes

Answer: (d) To remove toxic fumes — Ventilation removes toxic gases, fumes and dust from the workplace.

Q2. SHORT ANSWER QUESTIONS

a) Write significance of Therapeutic Window and how it differs from Therapeutic Index.

Significance: the therapeutic window is the range of plasma concentrations in which a drug is effective without harmful side effects. It tells doctors the right dose: a wide window means a safer drug with flexible dosing; a narrow window means the dose must be controlled and the blood level checked.

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Therapeutic window	Therapeutic index
A range of plasma concentrations (between MEC and MTC).	A ratio (number): $TD_{50} \div ED_{50}$.
Shows the concentrations at which the drug is effective without harm; guides dosing.	A quantitative measure of the safety of the drug.
Wide window = safer drug.	High TI = safer drug.

b) How structure of penicillin allows it to inhibit the transpeptidase enzyme?

The structure of penicillin resembles the amino acid sequence on which the cell-wall-strengthening enzyme (transpeptidase) normally acts. The enzyme mistakenly takes up penicillin; the strained four-membered beta-lactam ring then opens and binds permanently to the enzyme, inactivating it. Without a functional cell wall the bacteria burst and die.

c) What is the primary mechanism of action for opiate drugs, such as morphine, in providing analgesia (pain relief)?

Opiates bind to opioid receptors (mainly mu receptors) in the brain and spinal cord, like a key in a lock. This closes calcium channels and opens potassium channels, which hyperpolarises the neuron, so it cannot easily send the pain signal to the brain.

d) Why is it significantly more challenging to develop safe and effective antiviral drugs compared to antibacterial drugs (antibiotics)?

Bacteria are independent living cells with their own structures (e.g. cell walls) that a drug can attack without harming us. Viruses live inside our cells and use our own DNA/RNA machinery, so it is hard to kill the virus without harming healthy cells. Viruses also mutate quickly and can stay dormant in our genome, where present drugs cannot kill them.

e) What is the concept of pesticide resistance in a target pest population, linking it to the principles of natural selection and repeated application?

In a pest population a few individuals are naturally able to survive a pesticide. When the pesticide is applied, the susceptible pests die, while the resistant ones survive and reproduce, passing on their resistance (natural selection). With repeated application the whole population becomes resistant (immune), the pesticide becomes less effective, and more toxic pesticides have to be used.

f) Define soil salinization. How it can be caused by irrigation practices in regions experiencing higher temperatures and reduced precipitation.

Soil salinization is the build-up of soluble salts in the soil to a level that harms plant growth. Irrigation water contains dissolved salts. In hot regions the water evaporates quickly and leaves the salts behind in the top soil; with little rainfall the salts are not washed down, so they accumulate with each irrigation.

Soil salinization is not discussed in the chapter; this answer is from general knowledge.

g) How does the chemical industry contribute to both energy production from fossil fuels and the development of renewable energy technologies?

Fossil fuels: it refines crude oil by fractional distillation into gasoline, diesel and kerosene, and improves fuel quality by cracking and catalytic reforming (high-octane gasoline); natural gas is processed as a fuel. **Renewable energy:** it produces the materials needed for renewable technologies, such as pure silicon for solar panels, the chemicals for batteries, and hydrogen (from the chlor-alkali process and natural gas) as a clean fuel.

h) Name the industrial process used to produce caustic soda (NaOH) and chlorine (Cl_2) from the rock salt (NaCl).

The chlor-alkali process: electrolysis of sodium chloride brine (in membrane cells), which gives chlorine, sodium hydroxide and hydrogen.

i) State two different types of products (other than fuels) that are derived from the petrochemical industry, linking them to their use in daily life.

(1) Plastics, e.g. polyethene for bags, bottles and packaging. (2) Synthetic rubber for tyres and footwear (or synthetic fibres such as polyester for clothing).

j) Distinguish between a corrosive hazard and a toxic hazard, giving one example of a chemical for each.

Corrosive hazard	Toxic hazard
The chemical destroys living tissue (skin, eyes) and materials such as metals on contact.	The chemical poisons the body when inhaled, swallowed or absorbed, harming organs or causing death.
Example: concentrated sulphuric acid (or caustic soda).	Example: chlorine gas (or mercury, lead compounds).

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Drug A has an ED_{50} of 20 mg and a TD_{50} of 80 mg. Drug B has a Minimum Effective Concentration (MEC) of 5 mg/L and a Minimum Toxic Concentration (MTC) of 50 mg/dm^3. Drug C has a therapeutic window where the effective dose is 10 mg and the toxic dose is 15 mg.
(i) Define Therapeutic Index (TI) and calculate the TI for Drug A.
(ii) Define Therapeutic Window (TW) and identify which drug (B or C) has the wider therapeutic window. Justify your answer.
(iii) Compare the relative safety of Drug A and Drug C.

(i) The therapeutic index is a quantitative measure of a drug's safety; it compares the dose that causes a toxic effect in 50% of people with the dose that is effective in 50% of people.

$$\text{TI of Drug A} = TD_{50} \div ED_{50} = 80 \text{ mg} \div 20 \text{ mg} = 4$$

(ii) The therapeutic window is the range of drug concentrations in the plasma that are effective without causing harmful side effects (from MEC to MTC). Drug B: 5 to 50 mg/dm^3 (1 L = 1 dm^3), so the toxic level is 10 times the effective level. Drug C: 10 to 15 mg, so the toxic dose is only 1.5 times the effective dose. **Drug B has the wider therapeutic window.**

(iii) For Drug A the toxic dose is 4 times the effective dose (TI = 4); for Drug C it is only 1.5 times (15 ÷ 10). Drug A is therefore safer. Drug C has a narrow margin and would need careful dosing and blood monitoring.

b) (i) Contrast synthetic and organic fertilizers based on their origin.
(ii) Identify the three primary macronutrients (N, P, K) and describe the specific biological function of each in a plant.

(i) Synthetic fertilizers are man-made by chemical processes (e.g. the Haber-Bosch process for ammonia). Organic fertilizers come from natural sources such as compost or animal manure.

(ii) The primary macronutrients are nitrogen, phosphorus and potassium:

Nutrient	Function in the plant
Nitrogen (N)	Often the most limiting factor in crop production. A fundamental component of amino acids (which build proteins) and of chlorophyll, the pigment of photosynthesis.
Phosphorus (P)	Supplied as superphosphates or ammonium phosphates. A structural component of ATP, which stores and releases energy.
Potassium (K)	Does not become part of the plant's structure; acts as an osmoregulator and activates over sixty enzymes of plant metabolism.

c) Describe the chemical pathway from Natural Gas (CH_4) to urea fertilizer. Your answer must identify the crucial intermediate compound that is synthesized from CH_4 and air.

The crucial intermediate is **ammonia, NH_3**. Natural gas supplies the hydrogen and air supplies the nitrogen.

1. Methane reacts with steam to give hydrogen (and carbon dioxide).
2. Nitrogen from air and this hydrogen combine in the Haber-Bosch process (iron catalyst) to give ammonia.
3. Ammonia reacts with carbon dioxide to give urea.

$$CH_4 + 2H_2O \rightarrow CO_2 + 4H_2$$
$$N_2 + 3H_2 \rightleftharpoons 2NH_3 \text{ (Fe catalyst)}$$
$$2NH_3 + CO_2 \rightarrow NH_2CONH_2 + H_2O$$

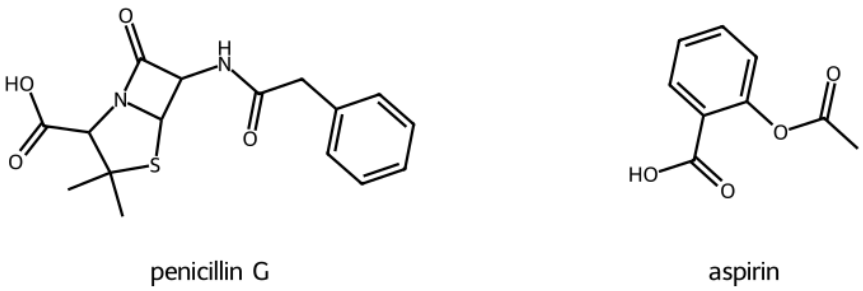
The chapter states that natural gas is the feedstock for ammonia, which is converted into urea, but does not give these equations; they are from general knowledge.

DESCRIPTIVE QUESTIONS

Q4. Describe the structure and mechanism of action of penicillin.

Structure

Penicillin contains two core rings: a four-membered **beta-lactam ring** and a five-membered **thiazolidine ring**, both carrying side chains. The beta-lactam ring is crucial for the antibiotic activity; it is also the site where bacterial enzymes called beta-lactamases can attack and deactivate penicillin.



Structures of a penicillin (penicillin G) and aspirin (own drawing)

Mechanism of action

1. The structure of penicillin resembles the amino acid sequence on which the bacterial enzyme that strengthens the cell wall (transpeptidase) normally acts.
2. The enzyme mistakenly takes up the penicillin molecule.
3. The beta-lactam ring opens and binds permanently to the enzyme, which is inactivated.
4. The bacteria continue to grow but without a functional cell wall, and eventually burst and die.
5. Human cells have no cell walls, so penicillin does not harm our cells.

Q5. What is Genetic Engineering? Give its main benefits and risks.

Genetic engineering in agriculture involves changing an organism's DNA by adding, removing or modifying specific genes. It is used to create genetically modified (GM) crops with desired traits.

1. **Identification:** scientists identify a useful gene with a desired trait (e.g. pest resistance from a bacterium, or drought tolerance).
2. **Isolation:** the gene responsible for the trait is isolated from the organism.
3. **Insertion:** the gene is inserted into the target plant's genome using a gene gun or a carrier bacterium.
4. **Expression:** the plant's cells express the new gene; the plant now produces a new protein and shows the new trait.

Benefits	Risks
Improved resistance to pests and diseases.	GM genes may spread to wild plants (outcrossing).
Tolerance to herbicides.	“Superweeds”: herbicide-resistance genes jump to wild weeds by cross-pollination.
Enhanced nutritional value, e.g. Golden Rice with beta-carotene to prevent blindness.	Resistant insects are produced.
Higher yields and less need for pesticides, e.g. BT-cotton makes its own pesticide.	Unknown long-term effects on human health and biodiversity.

Q6. Explain the role of industrial chemistry in economy of Pakistan. Also account the main raw materials available in Pakistan for these industries.

Role in the economy

- The chemical industry contributes significantly to the Gross Domestic Product (GDP) and provides jobs.
- It supports agriculture (fertilizers), construction (cement) and textiles (dyes and synthetic fibres).
- It uses locally available raw materials (natural gas, rock salt, limestone, crude oil), which reduces reliance on imports.
- Food security: local companies use natural gas to produce urea, cutting the import bill by billions of dollars and making fertilizer affordable.
- Textiles (over 50% of exports) consume caustic soda for mercerizing cotton, synthetic dyes for colour and hydrogen peroxide for bleaching.
- The cement industry has become a leading exporter to neighbouring regions.

Raw materials available in Pakistan

The feedstock of an industry determines its location and economic viability. Air provides nitrogen for fertilizers and oxygen for steel making; water is a solvent, a cooling agent and a source of hydrogen; fossil fuels provide the hydrocarbons for 90% of organic chemicals. Raw materials available in Pakistan:

Raw material	Use
Natural gas	Major source of methane (CH_4); used to produce fertilizers (urea) and petrochemicals; both a fuel and a source of hydrogen.
Rock salt (Khewra mines)	One of the world’s largest deposits; source of NaCl for soda ash (Na_2CO_3), caustic soda (NaOH) and chlorine.
Limestone	Abundant in the Salt Range and Sindh; source of $CaCO_3$ for the cement industry.
Gypsum	Cement industry and plaster; source of calcium and sulphur; soil conditioner.

Raw material	Use
Chromite	Found in Muslim Bagh; used in stainless steel and pigment production.
Crude oil	Refined to produce fuels and petrochemicals.
Sulphur	Used to make sulphuric acid (H_2SO_4).

Q7. Write a note on various industrial chemical processes. Also account for the applications of these chemical industries in Pakistan.

1. Polymerisation: small molecules (monomers) link to form large chains (polymers). In *addition polymerisation* monomers add to one another without losing any atoms (ethene → polyethene). In *condensation polymerisation* monomers join and release a small molecule, usually water (polyesters). Engro Polymer and Chemicals is the sole producer of PVC in Pakistan.

2. Refining and reforming: fractional distillation separates crude oil into gasoline, diesel, kerosene etc. by boiling point. Reforming converts straight-chain hydrocarbons into branched ones over a heated catalyst to give high-octane gasoline. Refineries such as PARCO and Attock Refinery supply petrochemical feedstocks.

3. Haber-Bosch process: ammonia is synthesised from nitrogen and hydrogen with iron as catalyst. It accounts for 13% of the nitrogen fixation on earth; 70% of ammonia goes into fertilizers. Engro Fertilizers and FFC use natural gas as feedstock and convert the ammonia into urea.

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \text{ (Fe catalyst)}$$

4. Contact process: SO_2 is converted into SO_3 using V_2O_5 or Pt as catalyst; SO_3 is changed into oleum and finally into highly pure, concentrated H_2SO_4, the "King of Chemicals". Example: Nimir Industrial Chemicals.

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} \ (V_2O_5)$$

5. Solvay process: produces sodium carbonate (Na_2CO_3) from brine and limestone. Ammonia is added to brine and CO_2 is passed to precipitate $NaHCO_3$, which on heating gives Na_2CO_3. Ammonia is regenerated using lime; the by-product is $CaCl_2$. Lucky Core Industries (formerly ICI) makes soda ash for glass, detergents and paper.

6. Chlor-alkali process: electrolysis of brine (NaCl) gives chlorine (Cl_2), sodium hydroxide (NaOH) and hydrogen (H_2), using modern membrane cells. The products are used for plastics, detergents and water treatment. Sitara Chemical Industries and Ittehad Chemicals are the primary producers.

$$2NaCl_{(aq)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + Cl_{2(g)} + H_{2(g)} \text{ (electrolysis)}$$

Applications

Industry	Role of industrial chemistry
Petrochemical	Crude oil is cracked into smaller molecules, the building blocks of plastics, synthetic rubber, fibres, lubricants and waxes.
Cosmetics	Makes ingredients for makeup, soaps and perfumes: surfactants (soaps), emulsifiers (smooth creams) and preservatives (prevent bacterial growth); products are pH-balanced for skin.
Cement	An intimate mixture of limestone and clay is heated to 1500 °C in a rotary kiln; the product is ground with gypsum. (Cement warms with water: exothermic hydration.)
Food	Additives and preservatives (sodium benzoate); hydrogenation of edible oils to solid fats like margarine; leavening agents (sodium bicarbonate); fortification (iodine in salt, vitamin A in oil).

Q8. What are important safety measures which must be followed while working in different chemical industries? Make a checklist to verify their adoption.

1. **Personal protective equipment (PPE):** gloves, goggles, helmets, lab coats and protective shoes to avoid injuries from chemicals, heat or harmful fumes.
2. **Labelling, handling and storage:** chemicals must be properly labelled, handled and stored according to their properties; flammable, toxic or corrosive substances are kept in suitable containers and separate storage areas to prevent leaks, spills, contamination or dangerous reactions.
3. **Emergency plans:** clear plans for fires, chemical spills, gas leaks or explosions; emergency exits, fire extinguishers, first-aid kits and spill-control materials must be available and workers must know how to respond.
4. **Ventilation:** good ventilation systems remove toxic gases, fumes and dust and keep the workplace safe and healthy.
5. **Training and maintenance:** workers must receive safety training on hazardous materials and equipment; regular inspection and maintenance of machines, pipelines and storage tanks detects faults early.

Checklist

Checklist item	Yes / No
Are all workers wearing the required PPE (gloves, goggles, helmet, coat, shoes)?	
Are all chemicals correctly labelled and stored in suitable containers?	
Are flammable, toxic and corrosive chemicals stored in separate areas?	
Is there a written emergency plan, and are exits clear and marked?	
Are fire extinguishers, first-aid kits and spill-control materials available?	

Checklist item	Yes / No
Is the ventilation system working?	
Have all workers received safety training?	
Are machines, pipelines and storage tanks inspected and maintained regularly?	

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 32.1

a) Drug A has a TI of 3, and Drug B has a TI of 50. Which drug is considered safer, and why?

Drug B is safer. A higher therapeutic index means a larger difference between the effective dose and the toxic dose, so there is less risk of toxicity. Drug A (TI 3) has its toxic dose close to its effective dose and needs careful monitoring.

b) How do antacids (e.g., magnesium hydroxide, $Mg(OH)_2$) chemically regulate stomach pH? Write a simple neutralization equation to illustrate this.

Antacids are weak bases. They neutralise the excess hydrochloric acid in the stomach, forming a salt and water, so the pH rises.

$$Mg(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$$

c) Identify the core, strained heterocyclic ring system in penicillin that is essential for its antibacterial activity.

The four-membered beta-lactam ring (fused to the five-membered thiazolidine ring).

d) Compare the chemical structures of morphine and codeine (methylmorphine). How does this small structural difference relate to codeine's action as a prodrug?

Both have the same ring skeleton. Morphine has a free phenolic –OH group on its benzene ring; in codeine this group is methylated, i.e. it is an $–OCH_3$ (methyl ether) group. Codeine itself binds weakly to opioid receptors. In the body (liver) the methyl group is removed, converting part of the codeine into morphine, which is the active pain-relieving form. A drug that is changed into its active form inside the body is called a prodrug.

The structures of morphine and codeine are not given in the chapter; this answer is from general knowledge.

e) Why do many antiviral drugs cause significant side effects for the host patient?

Viruses live inside our cells and use our own DNA/RNA machinery to replicate. A drug that stops the virus therefore also tends to interfere with the processes of healthy human cells, which causes side effects.

Quick Check 32.2

a) State two major agricultural benefits of using fungicides, linking to crop health or yield.

(1) They kill fungi that cause plant diseases, keeping the crop healthy and reducing the risk of epidemics. (2) By preventing damage and food loss they increase crop yield and contribute to a stable food supply.

b) Write the balanced chemical equation for reaction that occurs when powdered limestone ($CaCO_3$) is added to a lake or field to neutralize sulphuric acid (H_2SO_4) from acid rain.

$$CaCO_{3(s)} + H_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + H_2O_{(l)} + CO_{2(g)}$$

c) Identify two specific traits (e.g., heat tolerance, salt tolerance) that scientists aim to develop in new crop varieties to make them more resilient to climate change.

(1) Drought resistance (e.g. drought-resistant corn). (2) Flood tolerance (e.g. Sub1 rice, which survives two weeks under water). Heat tolerance (heat-tolerant wheat) is another.

Quick Check 32.3

a) Identify two major sectors of Pakistan's economy (other than agriculture) that are heavily reliant on the chemical industry, and name one chemical product essential for each.

(1) Textiles: caustic soda for mercerizing cotton (also synthetic dyes and hydrogen peroxide for bleaching). (2) Construction: cement.

b) Explain the physical principle of the process used to separate crude oil (in terms of intermolecular forces and boiling points) on which this separation depends.

Crude oil is separated by fractional distillation, which depends on differences in boiling point. Larger hydrocarbon molecules have stronger intermolecular (London dispersion) forces, so they have higher boiling points and condense lower in the column; smaller molecules have weaker forces and lower boiling points and rise higher before condensing.

c) Many cosmetics and perfumes use esters. Name the two types of reactants needed to synthesize an ester and name the catalyst typically used.

A carboxylic acid and an alcohol; the catalyst is concentrated sulphuric acid.

$$RCOOH + R'OH \rightleftharpoons RCOOR' + H_2O \text{ (conc. } H_2SO_4)$$

d) Explain the chemical reason why highly oxidizing chemicals (like concentrated HNO_3) must be stored separately from flammable organic chemicals (like ethanol).

A strong oxidising agent readily oxidises organic compounds, which act as fuels (reducing agents). If they come into contact, a fast, highly exothermic redox reaction can occur, which may cause a fire or an explosion.

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